MEC350 Fluid Mechanics

JetForce Studio - Course Mode Fallback Snapshot

Water Jet Impact Engineering Simulator

Generated	2026-08-04T12:00:00+00:00
Analysis	Steady two-dimensional control-volume momentum
Primary result	Force exerted by the fluid on the plate
Student name	[Complete before submission]
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Instructor / section	[Complete before submission]
Institution	[Complete before submission]

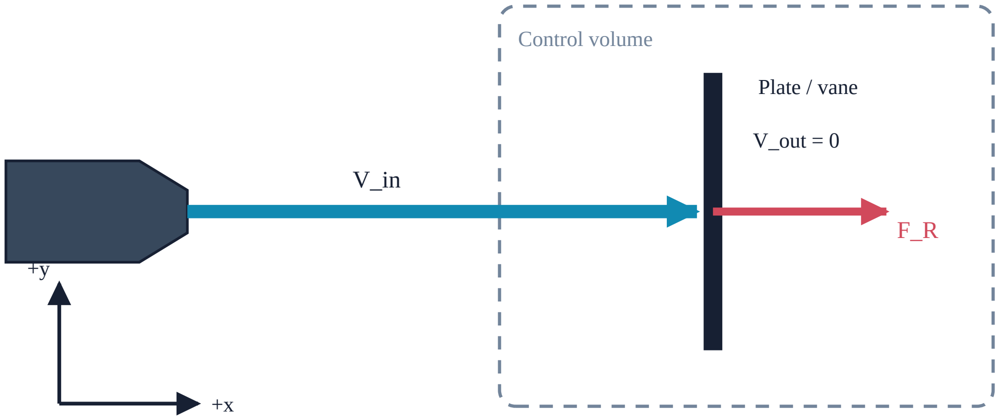
This application uses a numerical control-volume momentum model. The flow visualization is illustrative and is not a full CFD simulation.

Draft engineering report: the student must verify all inputs and results, complete the discussion, and add only references they have checked before submission.

1. Input parameters

Quantity	Value	Unit
Fluid density	1000	kg/m^3
Jet diameter	0.02	m
Inlet jet velocity	10	m/s
Impact model	normal_flat_plate	
Fluid preset	textbook_water	
Display unit preference	si	

2. Geometry schematic and control volume



Schematic - not to scale and not CFD

Angles are measured counterclockwise from positive x. Arrow lengths are normalized for legibility.

3. Assumptions

- Steady flow with no mass or momentum accumulation inside the control volume.
- Incompressible fluid with constant density.
- Uniform section-average velocity over each inlet and outlet jet.
- Exposed inlet and outlet free-jet surfaces are at atmospheric pressure, so gauge-pressure terms there are zero.
- The plate or vane is stationary and rigid.
- Gravity across the compact impact control volume and air resistance are neglected.
- The circular inlet jet retains its specified diameter before impact.
- The impact region is represented by a two-dimensional control volume.
- Non-ideal outlet loss is represented only by the prescribed velocity-retention coefficient k.
- The schematic and animation are engineering illustrations, not CFD solutions.

4. Governing equations

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A = pi d^2 / 4
Q = A V
mdot = rho Q = rho A V
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F_plate = mdot V_in - sum(mdot_j V_out,j)
Fx = mdot V_in,x - sum(mdot_j V_out,x,j)
Fy = mdot V_in,y - sum(mdot_j V_out,y,j)
FR = sqrt(Fx^2 + Fy^2)
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The momentum equation gives force on the fluid. The report displays the equal-and-opposite force exerted by the fluid on the plate.

5. Calculation procedure

1. Calculate circular jet area from the specified diameter.
2. Calculate volumetric and mass flow rates from the inlet speed.
3. Construct the model-specific outlet velocity vector or vectors.
4. Sum outlet momentum fluxes and subtract them from inlet momentum flux.
5. Report water-on-plate force components, resultant, and direction.

6. Main results

Quantity	Value	Unit
Jet cross-sectional area	3.141593e-04	m^2
Volumetric flow rate	0.00314159	m^3/s
Mass flow rate	3.14159	kg/s
Inlet velocity x-component	10	m/s
Inlet velocity y-component	0	m/s
Equivalent outlet velocity x-component	0	m/s
Equivalent outlet velocity y-component	0	m/s
Outlet momentum flux x-component	0	N
Outlet momentum flux y-component	0	N
Horizontal reaction force	31.4159	N
Vertical reaction force	0	N
Resultant reaction force	31.4159	N
Force direction angle	0	deg
Force direction defined	Yes	

7. Parametric-study figures

Force versus inlet velocity

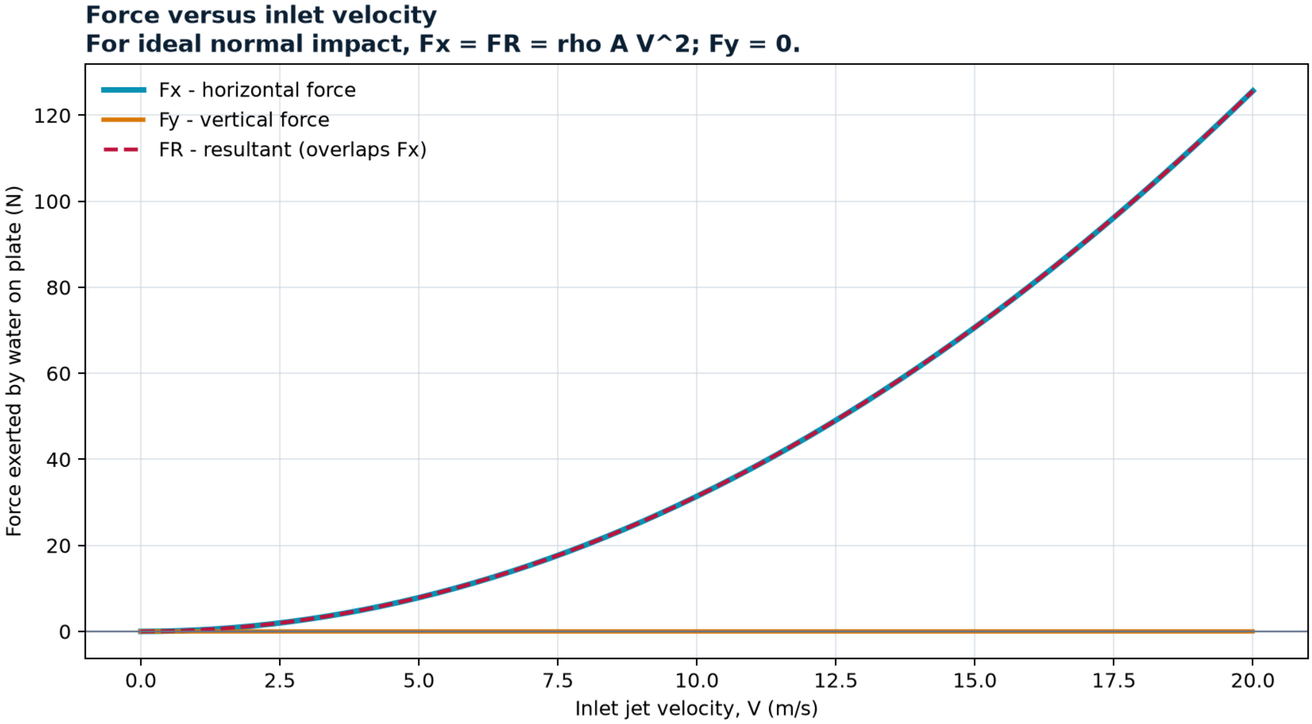

Figure 1. Force versus inlet velocity for the Course Mode normal flat-plate case. The plotted values come from the same control-volume momentum model as the live app.

Force versus jet diameter

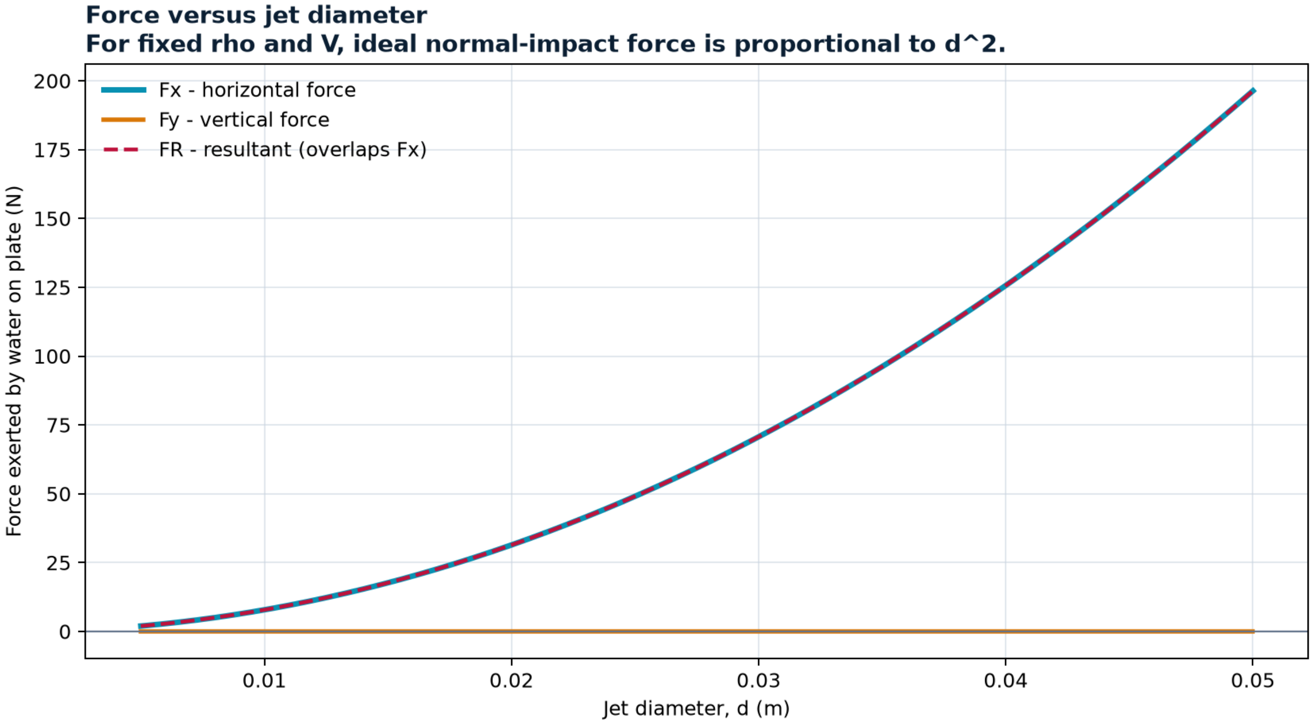

Figure 2. Force versus jet diameter for the Course Mode normal flat-plate case. The plotted values come from the same control-volume momentum model as the live app.

Table excerpt: first 7 of 122 parametric rows.

Study Variable	Inlet jet velocity [m/s]	Fx [N]	Fy [N]	FR [N]	Q [m^3/s]	mdot [kg/s]	Re [dimensionless]	Outlet speed [m/s]	Jet diameter [m]
velocity	0	0	0	0	0	0	0	0	Not applicable
velocity	0.333333	0.0349066	0	0.0349066	1.047198e-04	0.10472	6666.67	0.333333	Not applicable
velocity	0.666667	0.139626	0	0.139626	2.094395e-04	0.20944	13333.3	0.666667	Not applicable
velocity	1	0.314159	0	0.314159	3.141593e-04	0.314159	20000	1	Not applicable
velocity	1.33333	0.558505	0	0.558505	4.188790e-04	0.418879	26666.7	1.33333	Not applicable
velocity	1.66667	0.872665	0	0.872665	5.235988e-04	0.523599	33333.3	1.66667	Not applicable
velocity	2	1.25664	0	1.25664	6.283185e-04	0.628319	40000	2	Not applicable

8. Hand-calculation comparison

Quantity	Value	Unit
Given Data	rho = 1000 kg/m^3; d = 0.02 m; V = 10 m/s; normal flat plate	
Area Equation	A = pi d^2 / 4	
Jet cross-sectional area	3.141593e-04	m^2
Flow Equation	Q = A V	
Volumetric flow rate	0.00314159	m^3/s
Mass Flow Equation	mdot = rho Q	
Mass flow rate	3.14159	kg/s
Inlet Velocity Vector M S	(10, 0)	m/s
Equivalent Outlet Velocity Vector M S	(0, 0)	m/s
Momentum Equation	F_plate = mdot V_in - sum(mdot_out,j V_out,j)	
Horizontal reaction force	31.4159	N
Vertical reaction force	0	N
Resultant reaction force	31.4159	N
Analytical Verification	The scalar hand calculation and simulator agree within floating-point rounding because both apply the same documented analytical model.	

9. Discussion

Positive x follows the incoming jet and positive y is upward. The displayed force is the force exerted by the water on the plate.

For ideal normal impact, force is proportional to V^2. Doubling velocity from 10 m/s to 20 m/s produces four times the force.

This agreement is analytical verification, not independent experimental validation.

10. Conclusion

For rho = 1000 kg/m^3, d = 0.02 m, and V = 10 m/s, the ideal normal flat-plate model gives Fx = FR = 31.4159 N and Fy = 0 N.

11. Limitations

- The model does not solve spatial pressure, velocity, turbulence, or free-surface fields.
- Jet breakup, splashing, air entrainment, three-dimensional spreading, and nonuniform velocity profiles are not resolved.
- Transient loading, plate motion, torque, structural deformation, and fluid-structure interaction are outside scope.
- The velocity-retention coefficient is a user input unless supported by independently supplied measurements.
- Analytical equation checks are verification and are not experimental validation.

12. References

Placeholder - add only authoritative references personally checked by the student.

Student review required. Draft engineering report: the student must verify all inputs and results, complete the discussion, and add only references they have checked before submission.